Timothy Pham

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Education

University of Michigan – *Ann Arbor, MI* *Expected May 2027*
Master of Engineering in Space Engineering (GPA: 3.5/4.0)

University of Michigan – *Ann Arbor, MI* *May 2026*
Bachelor of Science in Aerospace Engineering

Skills

Programming: Python, MATLAB/Simulink, C++, C, VBA, HTML, Git, MBSE (SysML), SRIM, Mathematical Modeling
CAD/Simulation: Siemens NX (FEA), ANSYS, CST Studio, SPENVIS, CATIA, SolidWorks, AutoCAD, STK, Star-CCM+, XFoil
Hardware/Tools: Arduino, CAN Bus, LTSpice, AutoCAD Electrical, ETAS INCA, ESRI ArcMap, Power BI, Space Systems

Experience

Engineering Intern, *ND Defense – Madison Heights, MI* *May 2026 – Present*
- Validated tow bumper and chassis integrity across 12 projects using Siemens NX FEA to meet customer requirements
- Designed CAD fixtures for custom applications such as septic tanks, and integrated specialized electronics and hardware into center console and CAN bus systems across 4 projects
- Helping research ways to implement rear-mounted X-ray detection modules on COTS vehicle models
- Built an HTML-based document management system integrated with cloud infrastructure and custom dashboards, centralizing data for 500+ engineering projects
- Developed a quote automation tool that cut quote entry time by 80%, from 60 to 10 minutes per quote

Electrical Engineering Intern, *DTE Energy – Detroit, MI* *Jun 2025 – Dec 2025*
- Engineered city electrical grid conversion projects in ESRI ArcMap and AutoCAD Electrical, generating detailed work packages and circuit diagrams impacting 100K+ customers
- Orchestrated and implemented new transformers for coverage-wide deployment across Detroit, Michigan
- Collected field data and supported construction plan development in ESRI

Program Management Co-op, *DTE Energy – Detroit, MI* *Jan 2025 – Jun 2025*
- Supported design and commissioning of electrical systems across 15 concurrent infrastructure projects
- Authored work orders and project proposals that streamlined planning workflows and reduced approval cycle time
- Built a Power BI dashboard covering 150+ concurrent and 400 past projects to streamline project meetings

Vehicle Energy Efficiency Intern, *Hyundai Motor Group – Ann Arbor, MI* *May 2024 – Aug 2024*
- Utilized ETAS INCA to analyze vehicle performance data and calibrate fuel-economy parameters through CAN bus vehicle communication networks
- Optimized deceleration fuel cut-off (DFCO) strategies and transmission shift logic in latest ICE models, contributing to 18% increase in fuel efficiency
- Researched hybrid propulsion architectures and extended-range EV systems, synthesizing findings into technical reports for Korean engineering teams

Research

Research Systems Engineer, *Los Alamos National Laboratory Sponsored Research* *Aug 2026 – Present*
- Designing a small-satellite mission concept with LANL that uses a plasma contactor to raise local plasma density, shrinking the VLF loop antenna needed for whistler-wave generation from 200 m to 30 m
- Built Python models linking contactor plasma density to antenna radiation resistance and required transmit power, identifying a key density shortfall in baseline assumptions
- Simulated deployment of a 30 m loop antenna in MATLAB and performed FEA to verify structural stability against collapse after deployment
- Developed a power budget showing peak power drives battery sizing, and led platform, antenna, and orbit trade studies to support a flight proposal to LANL

Undergraduate Research Assistant, *University of Michigan – Ann Arbor, MI* *Sep 2025 – Apr 2026*
- Analyzed the effects of geomagnetic disturbances on spacecraft RF communication links to support ongoing space weather monitoring research
- Conducted aerospace vehicle simulations using STK and MATLAB scripts to evaluate ground station visibility and communication link budgets